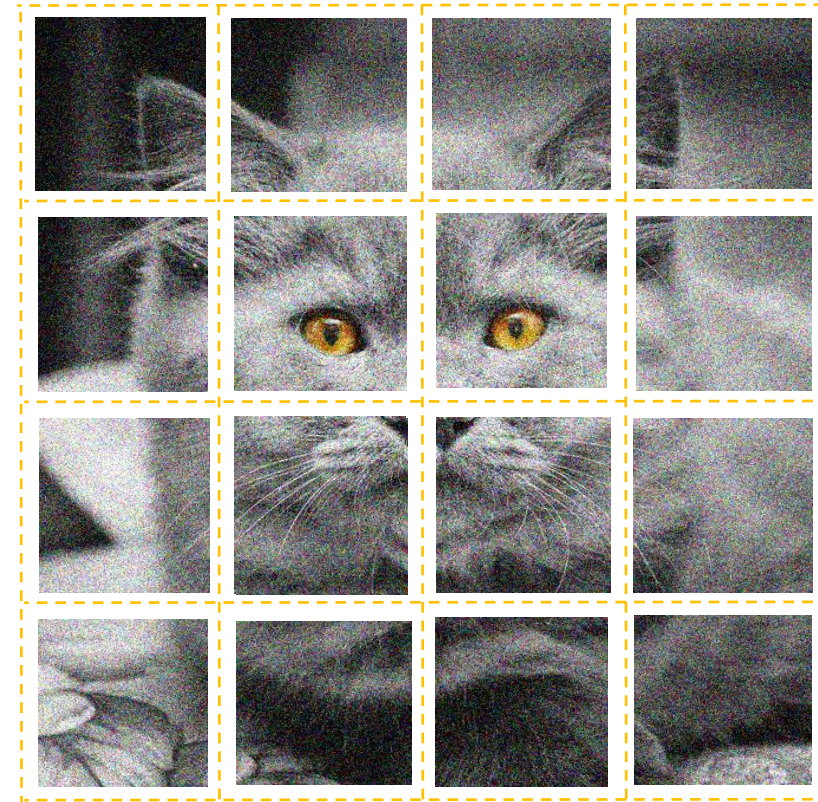
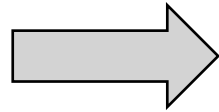
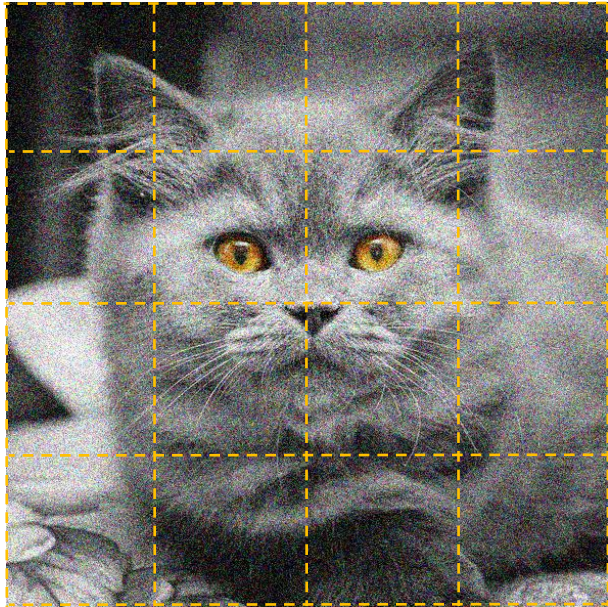


Noise2Void

- Self-supervised denoising – no clean images needed
- Blind-Spot Network – predicts pixels from context

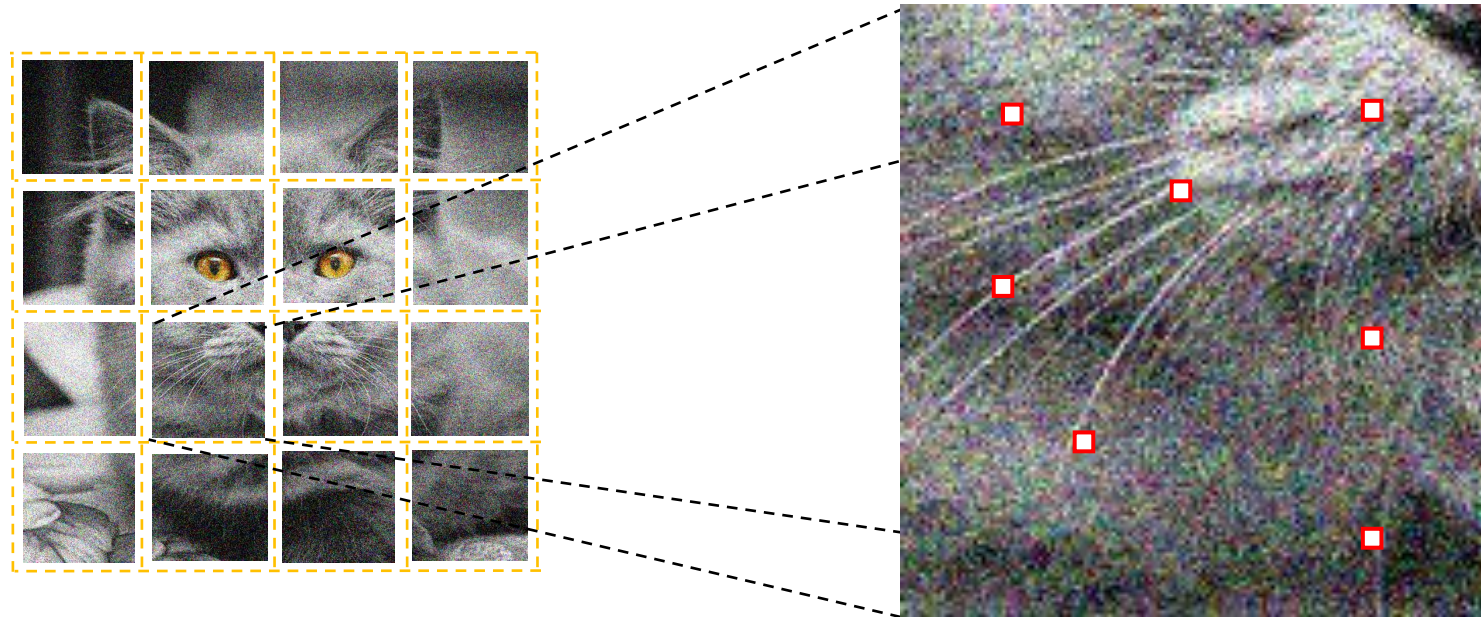
Method | Build Blind-Spot

1. Crop the image into small patches



Method | Build Blind-Spot

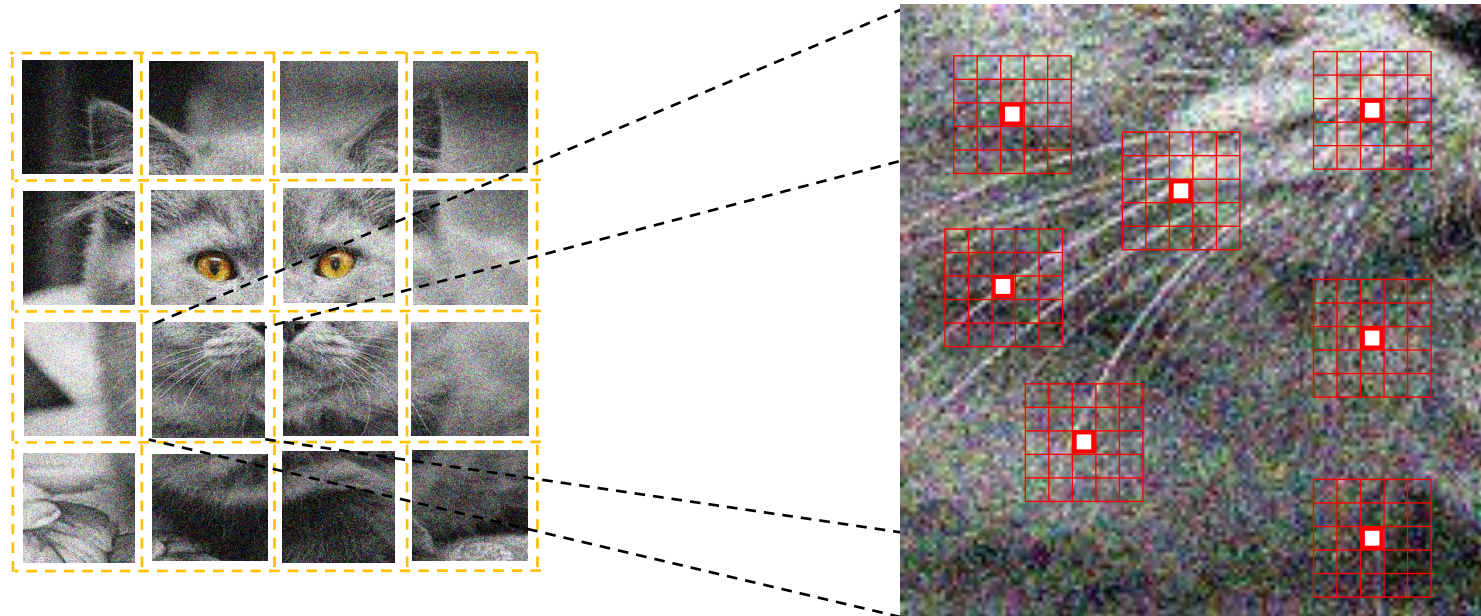
2. Inside each patch , mark $\approx 1-2\%$ of pixels as “blind spots”.



blind spots will be hidden from the network and used for the loss.

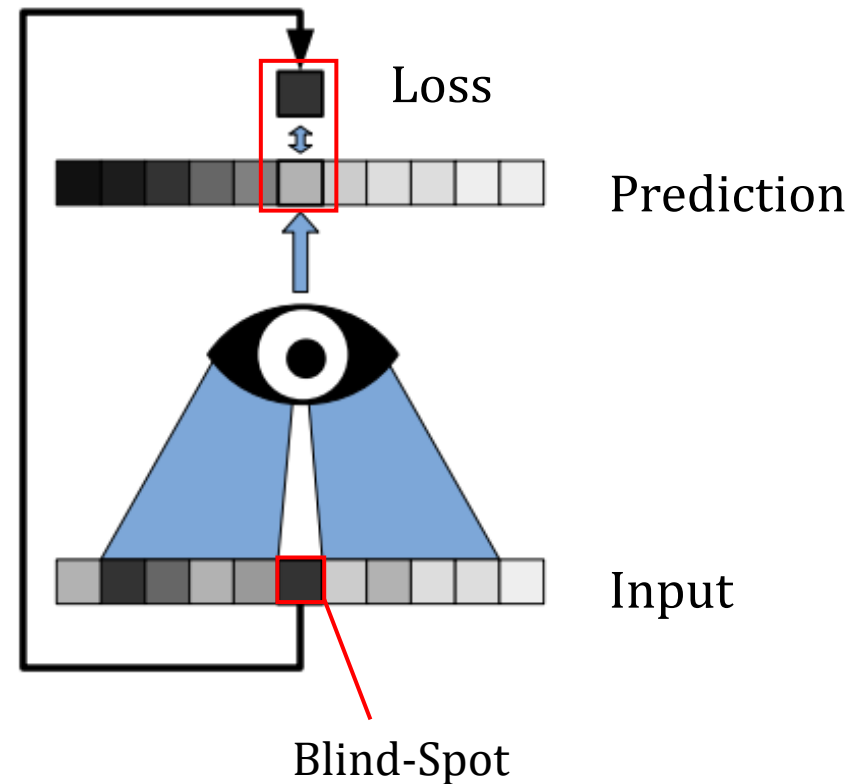
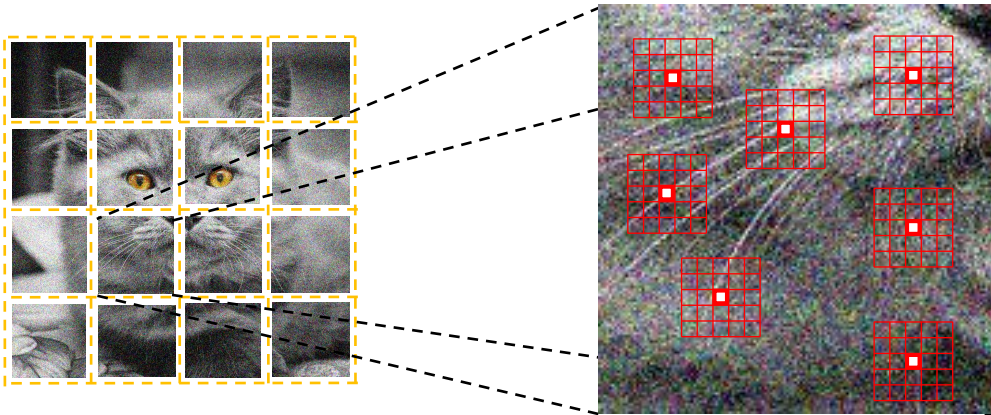
Method | Build Blind-Spot

3. For every masked pixel, copy a random neighbour (e.g. 5×5 window) into that location.



Method | Compute Loss & Train CNN

4. Use the blind-spot pixels and their predictions to compute the loss, then update the CNN.



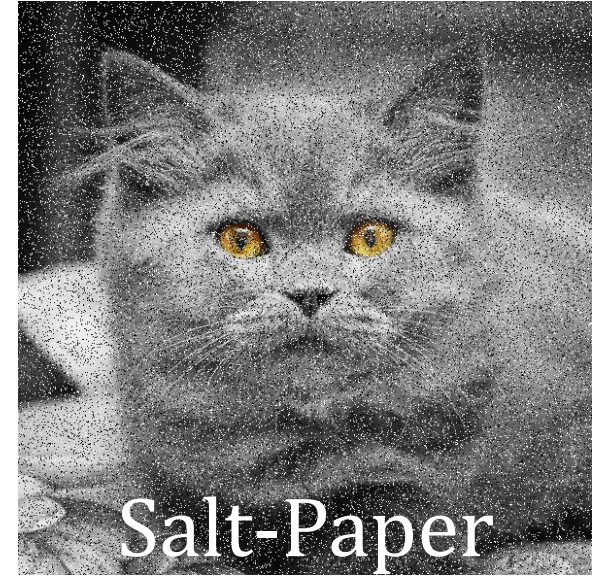
Introduction | Noise Types



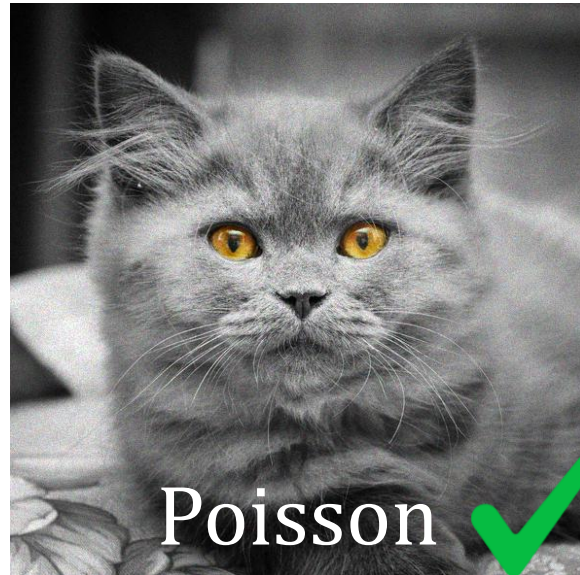
Original



Gaussian ✓



Salt-Paper

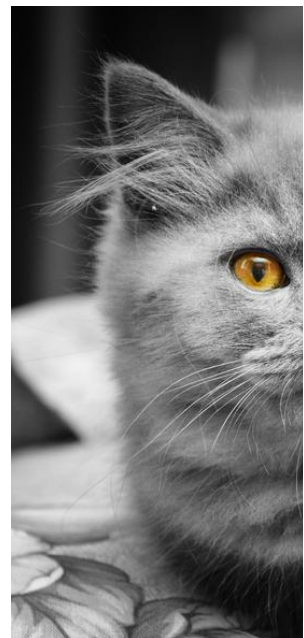


Poisson ✓

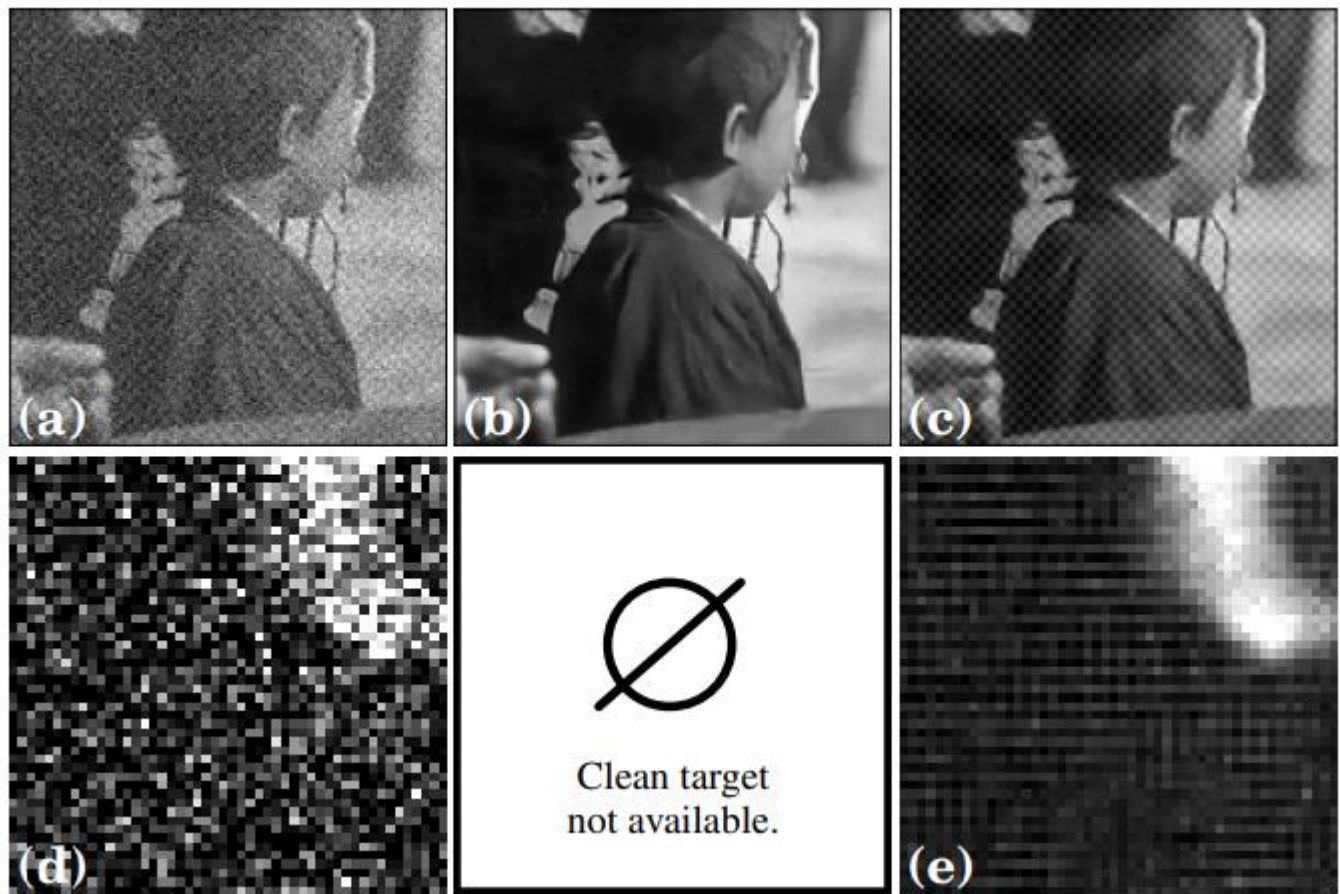


Speckle

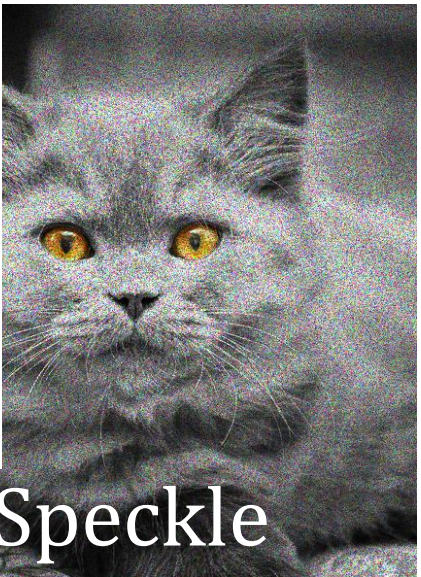
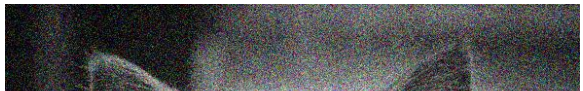
Introduction | Noise Types



Orig

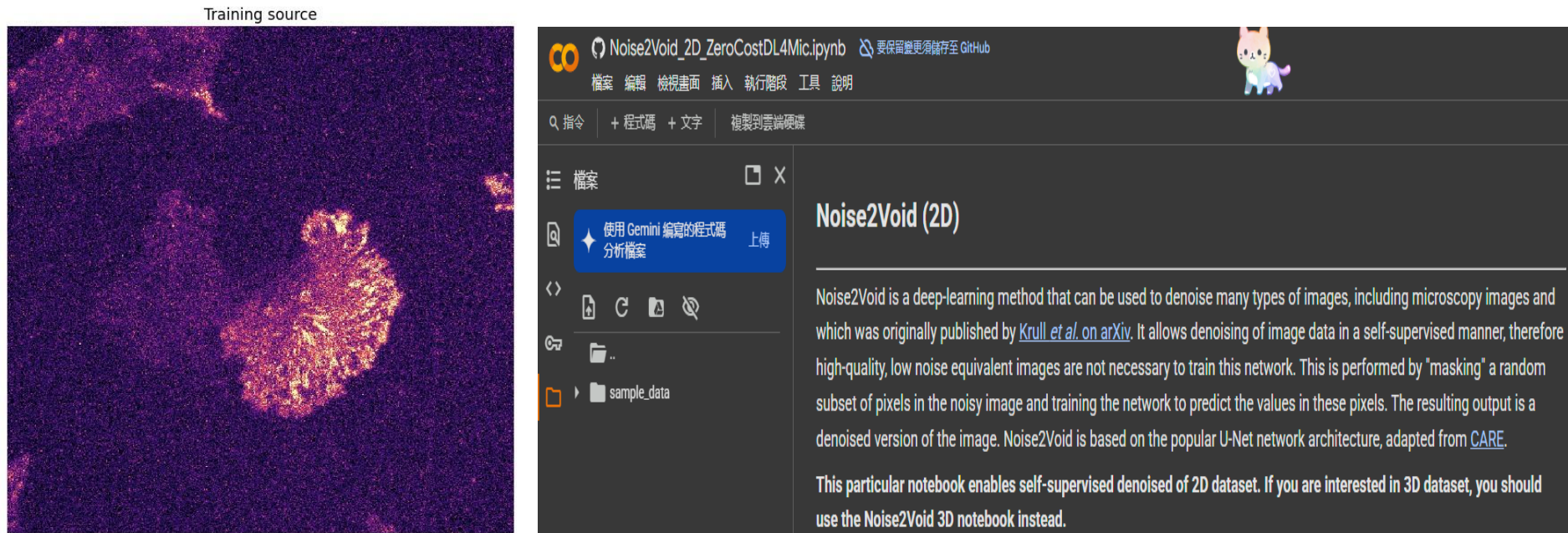


Poor Performance on structured noise



Implementation | Dataset and Environment Setup

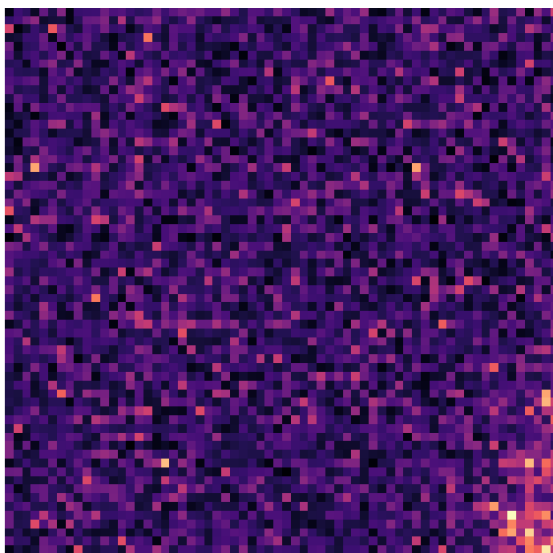
- Used the example image provided by the original N2V paper
- Platform: Google Colab + ZeroCostDL4Mic notebook
- One grayscale .tif image used for training



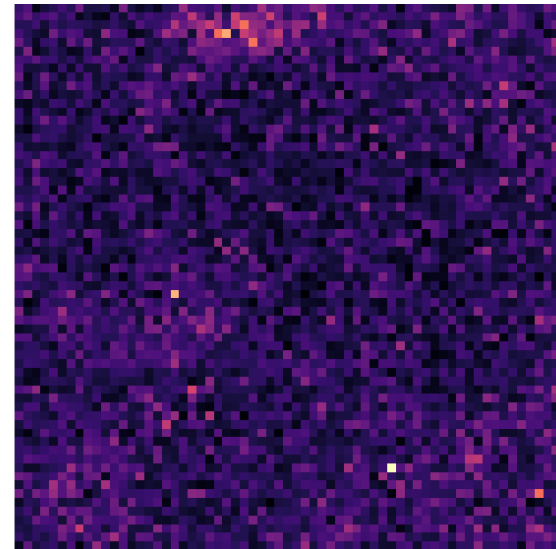
Implementation | Patch Extraction & Preprocessing

- Image automatically split into small patches: 64×64 pixels
- Patch count: 512 patches 90% for training (≈ 461) 10% for validation (≈ 51)
- Patches are normalized internally before training

Training Patch



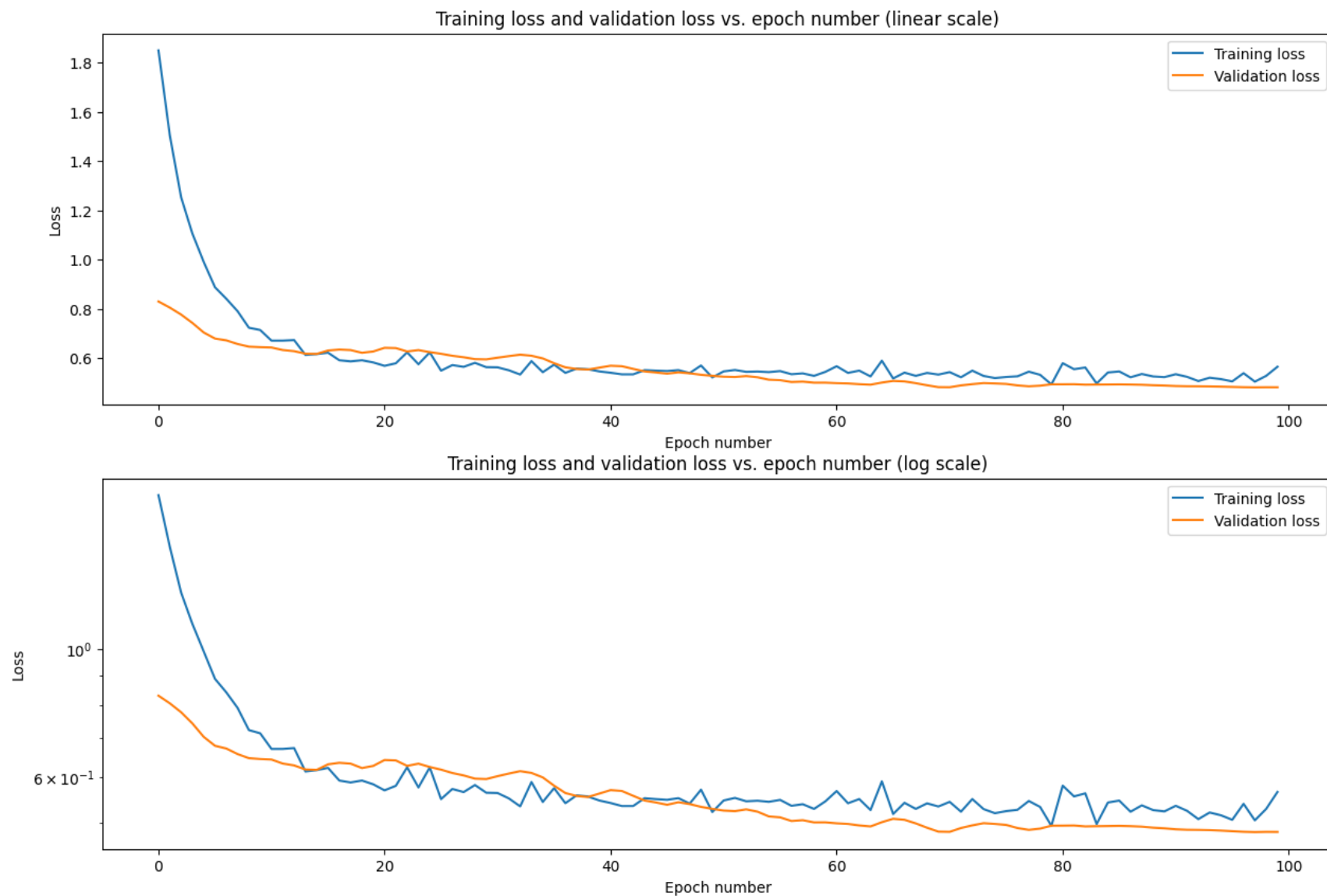
Validation Patch



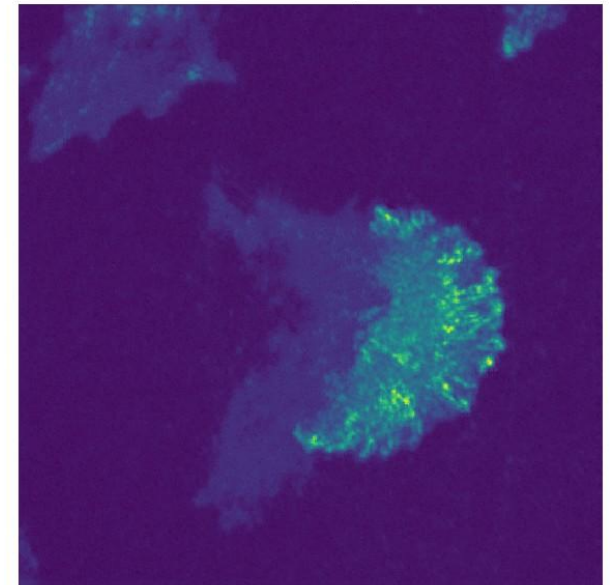
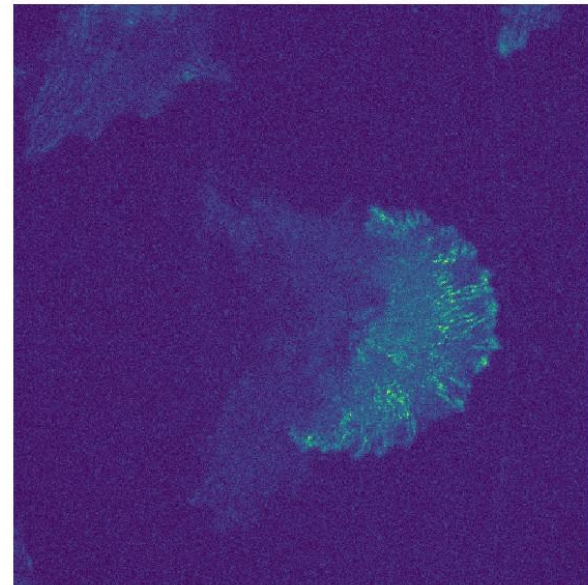
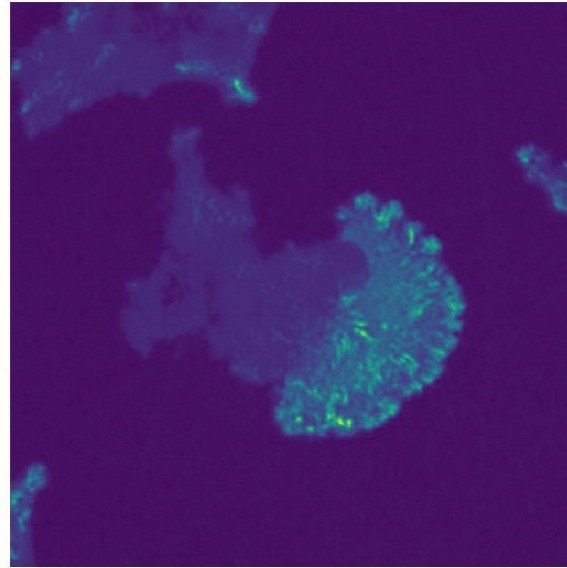
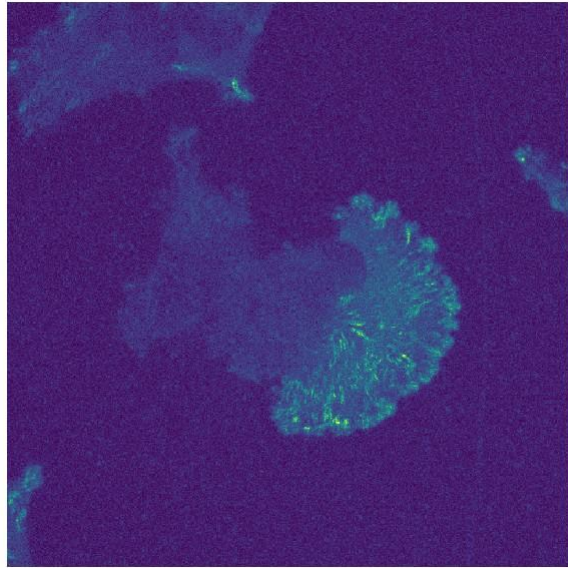
Implementation | Training Configuration

- Epochs: 100
- Steps per epoch: 4
- Batch size: 128
- Learning rate: 0.0004
- Loss function: Masked MSE
- Neighborhood radius: 5
- Blind-spot sampling: 19.8% pixels masked per patch

Implementation | Training Progress



Implementation | Denoising Results on Sample Image



Application | Chicken Coop Imaging

- Goal: Improve image quality inside poultry houses
- Challenge: Airborne dust and uneven lighting affect clarity



Application | Noise Reduction Results



Before



After

Application | Noise Reduction Results



Before



After

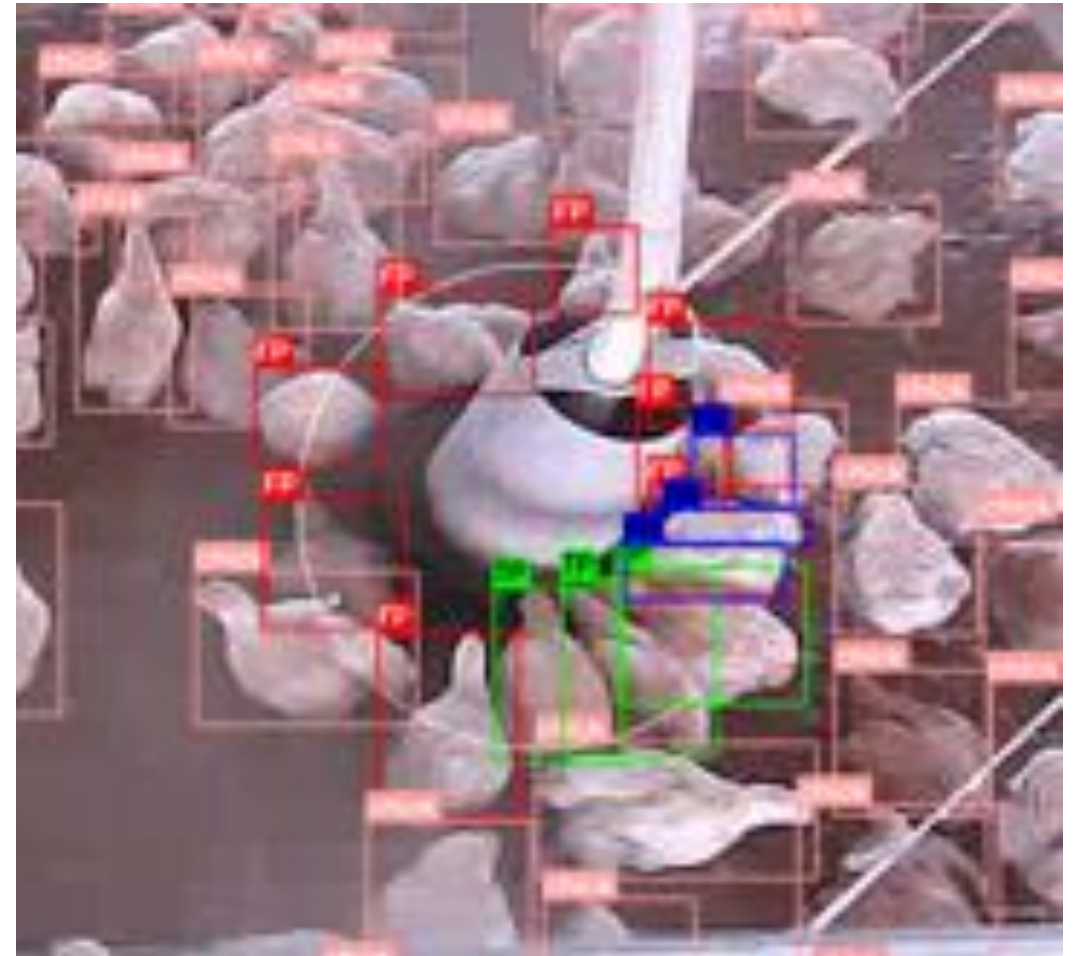
Application | Potential Benefits for Analysis

- Cleaner images may improve YOLO detection stability
- Fewer false detections from airborne dust or uneven lighting
- Could support more accurate feeder-zone chicken counting



Summary & Next Steps

- Trained N2V on real chicken coop images
- Reduced dust noise, clearer feeder/chicken visibility
- Next: test impact on YOLO detection
- Goal: integrate into feeding analysis pipeline



Reflection & Learning

-  Learned to train N2V from scratch with own data
-  Understood the importance of clean input for detection tasks
-  Balanced denoising and detail preservation
-  Bridged theory (deep learning) and field needs (agriculture)



Thanks For
Listening